

OIML R-76 METROLOGY LABORATORY

NON-AUTOMATIC WEIGHING INSTRUMENT (NAWI) VERIFICATION

Report Ref: R76-20260909-A87E12
Cert No: CERT-2026-DEL-00127
Date: 2026-09-09
Test Type: In-Service Verification

OVERALL METROLOGICAL
RESULT: PASS

Standard: OIML R-76-1:2006 (E)
Turning Point Method: Enabled (Small Weights)
Instrument meets all tested requirements of OIML R-76-1.

1. NAWI & METROLOGICAL SPECIFICATIONS

Manufacturer:	Essae-Teraoka Pvt. Ltd.	Accuracy Class:	Class III (Medium Accuracy)
Model / Type:	DS-15 Retail NAWI	Max Capacity (Max):	15 kg
Serial Number:	ET-2026-001527	Min Capacity (Min):	0.1 kg
Device Type:	Electronic Counter-Top Scale	Scale Interval (e):	0.005 kg
Customer / Location:	Sharma Retail Mart Pvt. Ltd. / Legal Metrology Laboratory, New Delhi	Actual Interval (d):	0.005 kg
Inspector / Verifier:	Er. R. Sharma (Senior Metrologist)	Divisions n = Max/e:	3,000 divisions

Temperature:	22 °C	Humidity:	52 % RH	Pressure:	1012.5 hPa	Weights Ref:	OIML Class M1 Standard Weights (1g - 20kg) (NABL-CAL-2026-00912)
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2. SUMMARY OF METROLOGICAL EVALUATIONS (OIML R-76-1)

OIML R-76 Clause / Evaluation Subject	Requirement / Criterion	Result
Instrument Metrological Limits (Table 3)	divisions n within class range; Min capacity verified	PASS
Zero Setting Accuracy (A.4.2)	Zero error $ E_0 \leq 0.25 e$	PASS
Tare Balancing & Net Weighing (A.4.6)	Net indication corrected error $\leq MPE$	PASS
Weighing Performance & Hysteresis (A.4.4)	Increasing & Decreasing load errors $\leq MPE$	PASS
Repeatability (A.4.10)	Span delta $(I_{max} - I_{min}) \leq MPE $	PASS
Eccentricity (A.4.7)	Errors at 5 load receptor positions $\leq MPE$	PASS

3. DETAILED METROLOGICAL TEST PROTOCOLS

Test Description	Nominal Load	Indication	Small Wt (ΔL)	Error (E_c)	MPE / Limit	Status
Zero Setting (A.4.2)	0.00 kg	0.0 kg	0.0025 kg	0.0 kg	$\pm 0.00125 \text{ kg}$ (0.25e)	PASS
Tare Balancing (A.4.6)	2.0 kg	2.0 kg	0.0025 kg	0.0 kg	$\pm 0.0013 \text{ kg}$	PASS
Tare Net Weighing (A.4.6)	Net: 5.0 kg	5.0 kg	0.0025 kg	0.0 kg	$\pm 0.01 \text{ kg}$	PASS

Eccentricity Test (clause A.4.7) — Test Load: 5.0 kg | MPE: $\pm 0.01 \text{ kg}$ | Max Diff: 0.002 kg

Pos #	Position Name / Description	Indication	Small Wt (ΔL)	Error (E)	Corrected (E_c)	Status
1	Center (1)	5.0 kg	0.0025 kg	0.0 kg	0.0 kg	PASS
2	Front-Left (2)	5.0005 kg	0.002 kg	0.001 kg	0.001 kg	PASS
3	Back-Left (3)	4.9995 kg	0.003 kg	-0.001 kg	-0.001 kg	PASS
4	Back-Right (4)	5.0 kg	0.0025 kg	0.0 kg	0.0 kg	PASS
5	Front-Right (5)	5.0 kg	0.0025 kg	0.0 kg	0.0 kg	PASS

Repeatability Test (clause A.4.10) — Requirement: $\Delta = (I_{max} - I_{min}) \leq |MPE|$

Load Level	Observed Indications (Runs 1..N)	I_max	I_min	Spread (Δ)	MPE	Status
7.5 kg	7.5, 7.5, 7.505	7.505 kg	7.5 kg	0.005 kg	± 0.01 kg	PASS
15.0 kg	15, 15.005, 15	15.005 kg	15.0 kg	0.005 kg	± 0.015 kg	PASS

Weighing Performance Test (clause A.4.4) — Max Hysteresis: 0.0 kg

Load (L)	I (Inc)	Ec (Inc)	I (Dec)	Ec (Dec)	Hysteresis	MPE ($\pm$)	Status
0.1 kg	0.1 kg	0.0	0.1 kg	0.0 kg	0.0 kg	± 0.005 kg	PASS
2.5 kg	2.5 kg	0.0	2.5 kg	0.0 kg	0.0 kg	± 0.005 kg	PASS
5.0 kg	5.0 kg	0.0	5.0 kg	0.0 kg	0.0 kg	± 0.01 kg	PASS
10.0 kg	10.0 kg	0.0	10.0 kg	0.0 kg	0.0 kg	± 0.01 kg	PASS
15.0 kg	15.0 kg	0.0	15.0 kg	0.0 kg	0.0 kg	± 0.015 kg	PASS

Tested / Verified By:	Approved / Authorized By:	Official Seal & Verification Date:
		Date: 2026-09-09
Technical Metrologist / Inspector	Quality Manager / Signatory	Place: Legal Metrology Laboratory, New Delhi
		Ref: R76-20260909-A87E12

METROLOGICAL NOTICE & STATUTORY DISCLAIMER:
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